

Name of the Student	V Yogeswar
Reg. No	192425175
GitHub Link	https://github.com/Yogeswar2906/MLA0405---DEEP-LEARNING/tree/main/Assessment/CO4

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

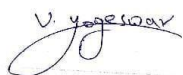
Course Outcome Weightage: 10%
Assessment Tool Weightage: 40%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I V Yogeswar / 192425175 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

V Yogeswar
Name of the student:

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

QUESTIONS: 05

Total Marks:25

Q1. Transfer Learning

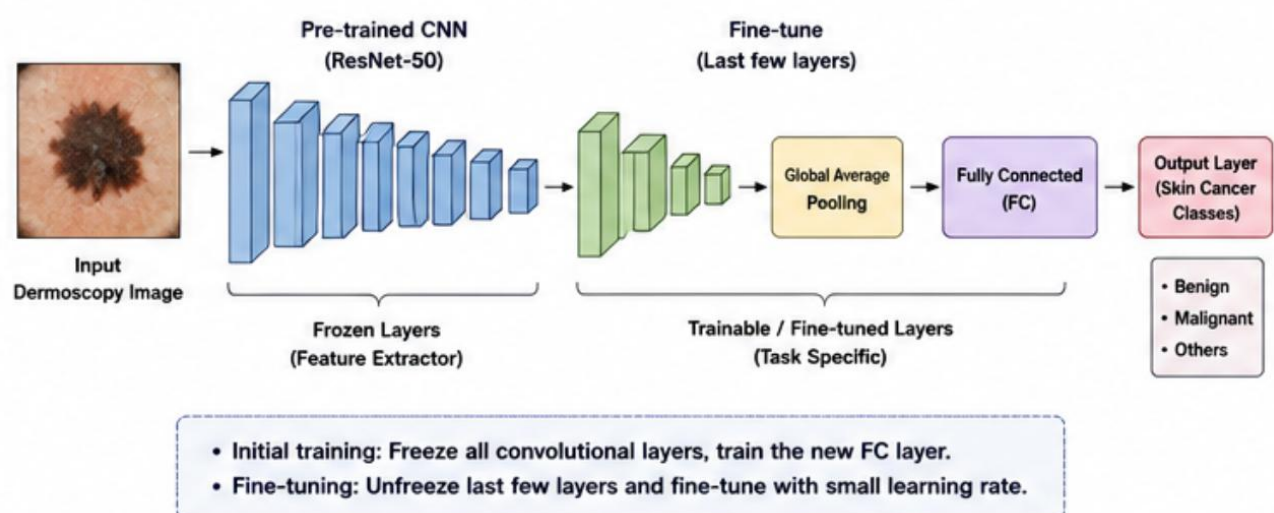
A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Proposed Strategy

Dermoscopy Image → Pre-trained CNN → Feature Extraction → Fully Connected Layer → Skin Cancer Classes

1. Resize and normalize the dermoscopic images to the input size required by the pre-trained model.
2. Load the pre-trained ResNet-50 model without its original classification layer.
3. Initially freeze the convolutional layers so that the learned general visual features are preserved.
4. Replace the original classification layer with a new layer suitable for the skin cancer classes.
5. Train the new classification layer using the available labeled images.
6. Then unfreeze the last few convolutional layers and fine-tune them using a small learning rate.

This approach is suitable because the dataset contains limited labeled images. Freezing most layers reduces training time and the risk of overfitting, while fine-tuning the deeper layers allows the model to learn skin-lesion-specific features. Therefore, transfer learning can provide better classification performance with less labeled data.



Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational

efficiency.

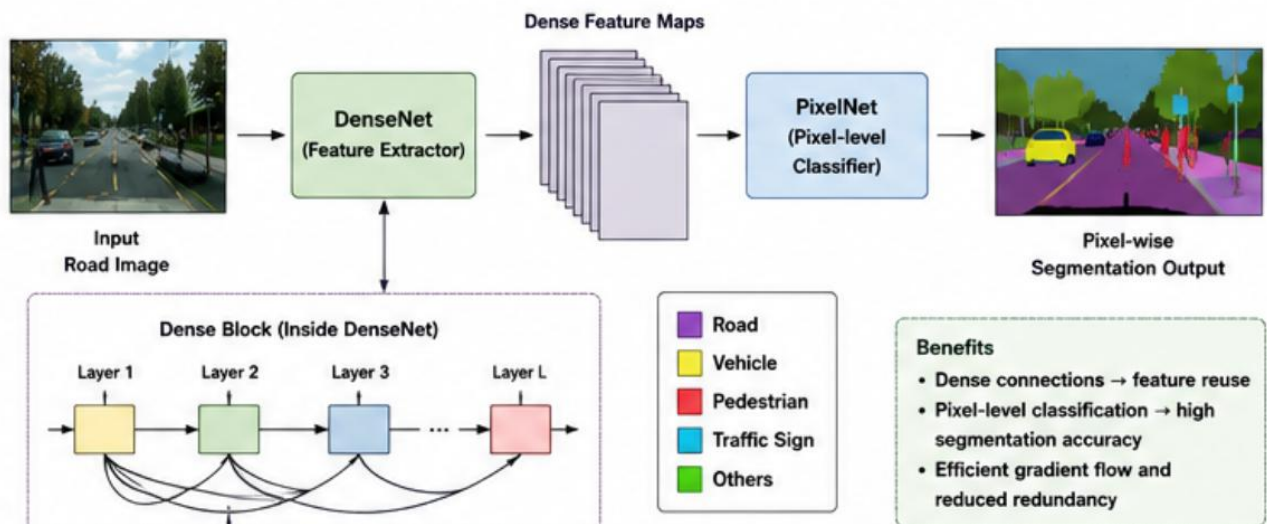
Proposed Architecture

Road Image → DenseNet Feature Extractor → Dense Feature Maps → PixelNet Pixel-Level Classifier → Segmentation Map

- **DenseNet:** The convolutional layers are connected densely, so each layer receives feature maps from previous layers. This encourages feature reuse and improves gradient flow.
- **PixelNet:** Uses learned pixel-level features to classify individual pixels into categories such as road, pedestrian, vehicle and traffic sign.
- **Output:** A pixel-wise segmentation map in which every pixel is assigned to a particular class.

Dense connections reduce redundant feature learning because previously learned features are reused by later layers. Pixel-level classification provides accurate object boundaries and detailed segmentation. The reuse of features can also improve parameter efficiency compared with learning completely separate features at every layer.

Thus, the combination of DenseNet for strong feature extraction and PixelNet-style pixel-level prediction is suitable for accurate road-scene semantic segmentation.



Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Architecture

Input Sentence → Embedding → Bidirectional RNN Encoder → Context Representation → Decoder → Translated Sentence

1. The input words are converted into numerical vectors using an embedding layer.
2. The Bidirectional RNN processes the input sequence in both forward and backward directions.
3. The forward RNN captures information from the past, while the backward RNN captures information from the future.
4. Their outputs are combined to form a context representation containing information from the

complete input sentence.

5. The decoder generates the translated sentence one word at a time.

Bidirectional processing is useful because the meaning of a word can depend on both preceding and following words. Therefore, the encoder obtains richer contextual information, which can improve translation accuracy, especially for multilingual and context-dependent sentences.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

An LSTM-based stock prediction system can be designed as:

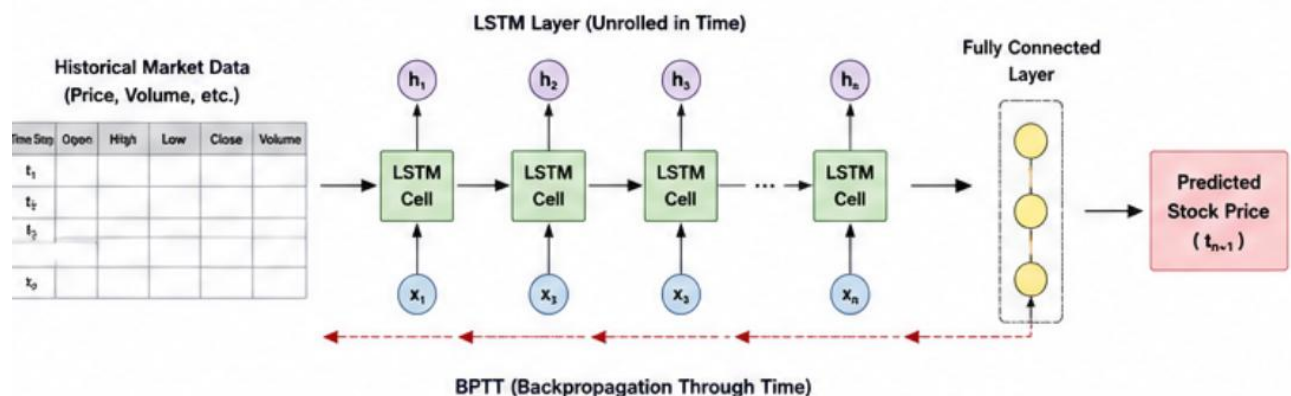
Historical Stock Data → Preprocessing → LSTM Layers → Fully Connected Layer → Predicted Stock Price

The historical data may contain features such as opening price, closing price, high, low and trading volume.

Working of LSTM and BPTT

1. Historical data is divided into sequences of time steps.
2. The LSTM receives these sequences and maintains information using its cell state and gates.
3. The LSTM output is passed to a fully connected layer to predict the future stock price.
4. During training, the prediction error is calculated using a suitable loss function such as Mean Squared Error.
5. Backpropagation Through Time (BPTT) unfolds the recurrent network across the time steps and propagates the error backward through these steps.
6. The weights are updated using gradient descent.

LSTM is more suitable than a traditional RNN because it uses input, forget and output gates to control information flow. Its cell state helps preserve important information over long sequences and reduces the vanishing-gradient problem. This makes LSTM better suited for learning long-term dependencies in historical financial data.



Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Proposed Architecture

Video → Frame Extraction → Pre-trained CNN → Visual Feature Vectors → LSTM Encoder → LSTM Decoder → Generated Caption

Working

1. The uploaded video is divided into frames.
2. A pre-trained CNN such as ResNet extracts visual features from each frame.
3. Transfer learning allows the CNN to provide useful visual representations without training the entire CNN from scratch.
4. The sequence of visual feature vectors is provided to an LSTM encoder, which captures the temporal information in the video.
5. The LSTM decoder generates the caption word by word.
6. The decoder continues generating words until an end-of-sentence token is produced.

Roles of Components

- Transfer Learning/CNN: Extracts meaningful visual features from video frames.
- LSTM Encoder: Learns the temporal relationships between frames.
- LSTM Decoder: Converts the learned visual-temporal representation into a natural-language caption.

The complete system combines spatial feature learning from CNNs with temporal sequence learning from LSTMs. Therefore, it can understand both the visual content and the sequence of events in a video, enabling accurate and meaningful automatic caption generation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context

Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis